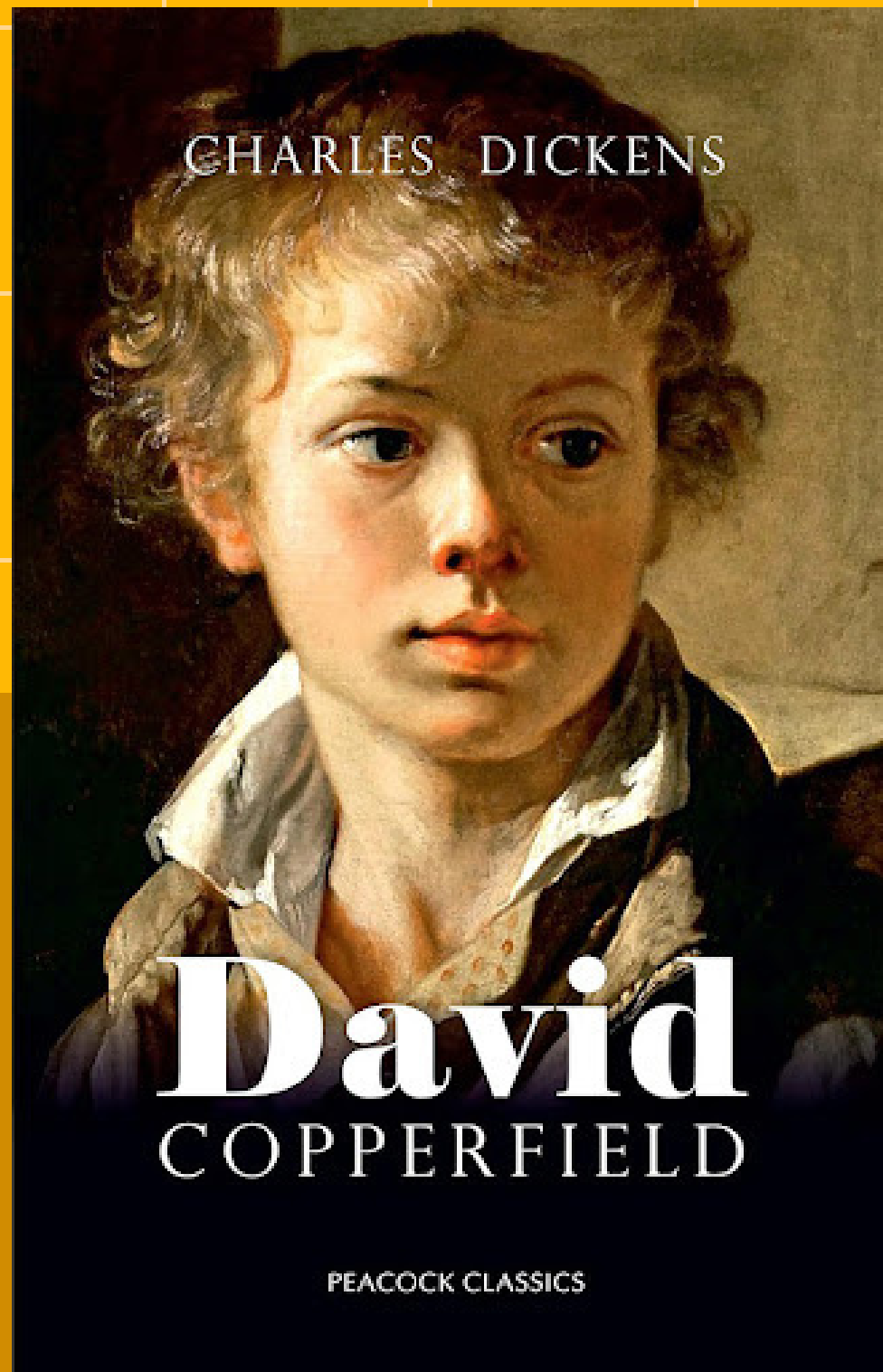


# NEXT WORD PREDICTION

Parmis Davari, Narges Elyasi,  
Moho Barabadi





# DAVID COPPERFIELD

- A classic coming-of-age novel
- Follows the life of its titular character from childhood to maturity, chronicling his struggles, friendships, and triumphs.
- Blends autobiography with social commentary.
- Dickens himself called it his "favorite child,"



# INTRODUCTION

- Providing a rich linguistic foundation for training the next-word prediction model.
- Reflecting the book's complexity and vocabulary diversity.

Analyzing this classic text allows the model to learn Dickens' distinctive style, enhancing its predictive accuracy.

**14.81K**

Number of Sentences

**439K**

Total Number of Words

**16.88K**

Number of Unique Words



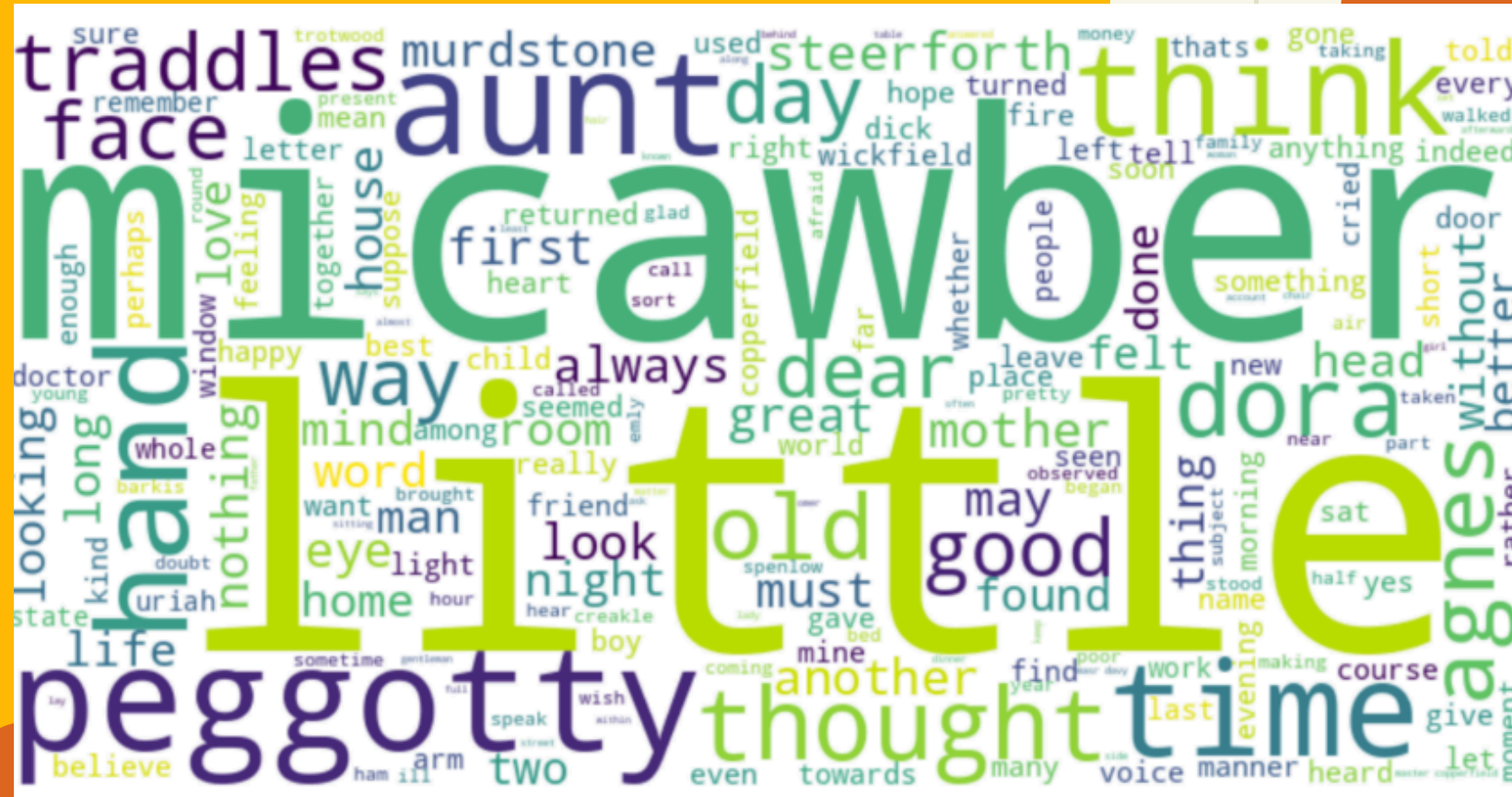
# WORD CLOUD

## Our Vision

- Highlighted frequently used words such as **"little", "time", "know", "mother", "think",** and **"aunt"**.
- Reflecting Dickens' emphasis on **family relationships, personal growth,** and the **passage of time.**

## Our Mission

## Make a predictive model with help of Dickens's rich book



# POS DISTRIBUTION

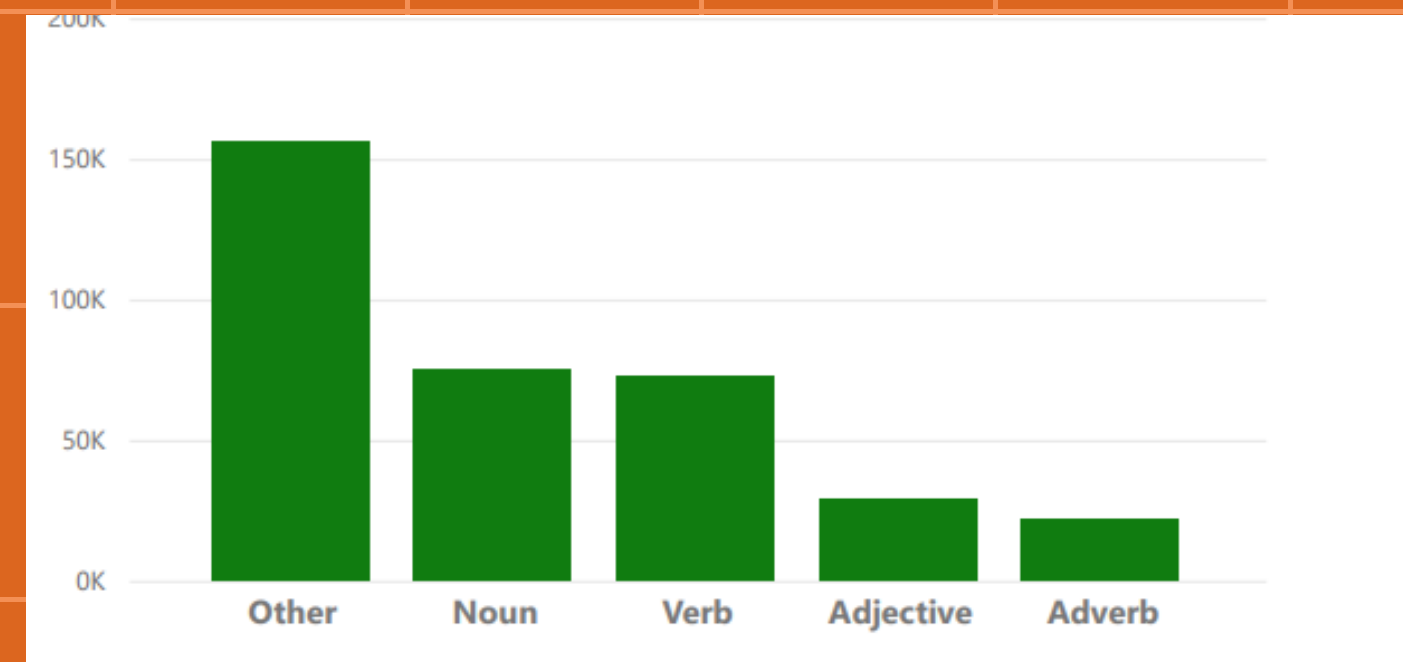
## POS TRANSITIONS

Nouns and verbs > Adjectives and adverbs

| From_POS  | Adjective | Adverb | Noun  | Other | Verb  |
|-----------|-----------|--------|-------|-------|-------|
| Verb      | 5808      | 8338   | 6075  | 41410 | 11035 |
| Other     | 16575     | 6225   | 41991 | 57413 | 33975 |
| Noun      | 1299      | 3850   | 9729  | 42101 | 17984 |
| Adverb    | 3852      | 2663   | 652   | 7971  | 6788  |
| Adjective | 1858      | 850    | 16516 | 7283  | 2884  |

### POS Transitions

Highlights transitions ending in  
**Noun** and **Verb**



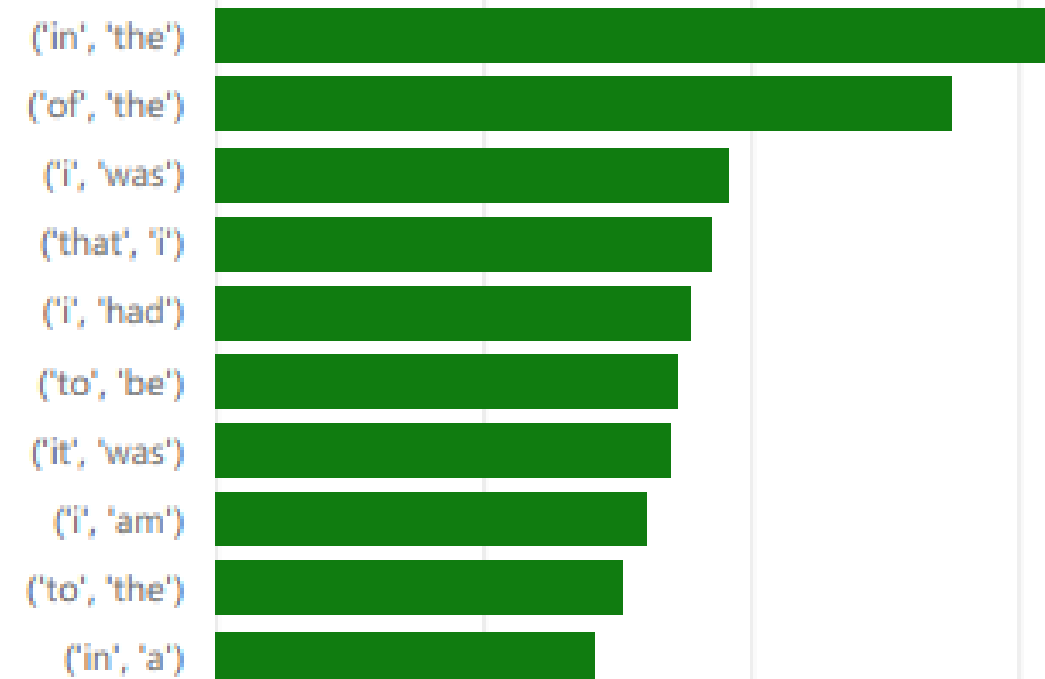
### POS Distribution

Focus on **action** and  
**character development**  
rather than excessive  
description.

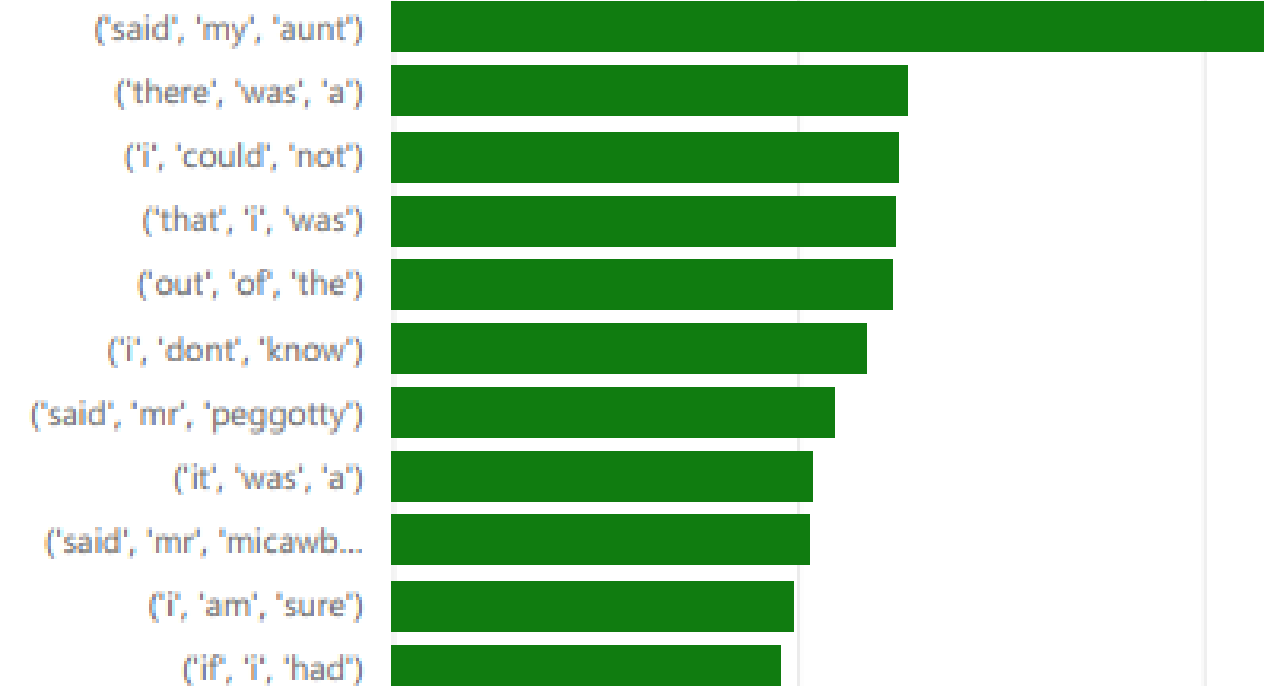




# BIGRAMS-TRIGRAMS



Top 10 Bigrams



Top 10 Trigrams

The prominence of "said my aunt" mentions the notable aunt characters, Betsey Trotwood.



# MODEL: LLAMA 3.2 INSTRUCT

## Training Pipeline:

- 1.Fixed Dependencies with **Poetry**
- 2.Loaded the Data (Sentences from the Database)
- 3.Trained with **Llama** and **Unsloth**

## Test Pipeline:

- 1.Loaded the Trained Model
- 2.Tested
- 3.Evaluate metrics



# TRANSFORMER/ LSTM

## Transformer Output

Generated text: **my deepest sorrows are** wings that she should come to see her house ; but she hurt bodily and all to me . ” “ i cannot leave her , ” said i , “ bear more than i can count , if i had had been brought up stronger than ever since . if i have been unhappy , i have loved her promise nothing , and the gentle cheerfulness way of doing anything but it — in the <unk> ! ” “ oh , my goodness gracious sake , trotwood , ” i said , “ you ’ ll pray for me . if i couldn ’ t touch her ! ” “ i am very miserable , ” said i , “ and i never was of it

## LSTM Output

Generated from **My deepest sorrows are**  
'my deepest sorrows are not very much attached it to see the little <OOV> of my <OOV> take possession of the <OOV> of my <OOV> that <OOV> like, for another one of her lips, and she was a small outside table i am sure, but to <OOV> the subject of her <OOV> and to i told you, i should wish to feel it made the money into the <OOV> of my aunts having been sometimes to <OOV> he made a <OOV> of the fire of a <OOV> and the <OOV> <OOV> <OOV> the <OOV> <OOV> of mr jack <OOV> which was a too, and he had gone to mr peggotty, and mr i told him well, and i could <OOV> my throat, to be got i the time when i first came'







# LLAMA RESULTS

**my deepwst sorrows are** soothed by the sight of you, and my heart is lifted when I see your lovely face.\nYou are a shining light in a world filled with darkness, a beacon of hope in times of despair. Your presence calms me, and brings peace to my soul.\n\nI am forever changed by our time together, and I will carry the memories of our moments shared with me always. You have shown me that even in the most trying of times, there is beauty to be found, and love to be experienced.\n\nThank you for being my guiding star, my safe haven, and my forever home. I promise to cherish and honor you



# THANK YOU

